Fiscal Year:	FY 2010	**Task Last Updated:**	FY 05/05/2010
PI Name:	Durian, Douglas J. Ph.D.		
Project Title:	The Melting of Aqueous Foams		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-6396	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	1995 Microgravity Fluid Physics NRA-94-OLMSA-05: Research and Flight Experiment Opportunities
Start Date:	08/05/2004	**End Date:**	04/04/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Motil, Brian J.	**Contact Phone:**	216-433-6617
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Flight Program:			
Flight Assignment:	NOTE: Grant number was changed in 2007 to NNX07AP20G (Ed., 1/4/13) NOTE: Funding extended to 4/04/2013 per NSSC database (5/2010) NOTE: Funding extended to 4/4/2010 through a new grant in 2007 (6/2008) NOTE: Funding reinstated thru 9/30/2008, per F. Chiaramonte (8/2006). Suppl funding thru Oct 2006 (6-5-06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX07AP20G ; NNX07AK60G ; NNC04GB61G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The unusual elastic character of foams will be quantified macroscopically by measurement of the shear stress as a function of static shear strain, shear strain rate, and time following a step strain; such data will be analyzed in terms of a yield stress, a static shear modulus, and dynamical time scales. Microscopic information about bubble packing and rearrangement dynamics, from which these macroscopic non-Newtonian properties presumably arise, will be obtained non-invasively by novel multiple-light scattering diagnostics such as diffusing-wave spectroscopy (DWS) and speckle-visibility spectroscopy (SVS). Bubble size and homogeneity will be studied by diffusely transmitted light (DTS). Liquid fraction will be measured by electrical conductivity. Quantitative trends with materials parameters, such as average bubble size, and liquid content, will be sought in order to elucidate the fundamental connection between the microscopic structure and dynamics and the macroscopic rheology. There are three all-encompassing objectives motivating the FOAM project. The first concerns multiphase flows and represents the “mechanics” aspect alluded to in the F.O.A.M. acronym: To understand how foams and bubbly liquids deform and flow in response to applied forces. The second objective concerns multiphase separations: To understand how foams and bubbly liquids evolve with time via a combination of liquid flow between bubbles, of gas diffusion across the film separating neighboring bubbles, and of direct coalescence via film rupture. For both flows and separations, some level of understanding has been established for very dry foams, and also for liquids with relatively few bubbles; however, we are more ignorant of the crucial intermediate regime of very wet foams and very bubbly liquids. This regime defies earth-bound experiments. The third, overall, objective is to transfer this new understanding of multiphase flows and separations into ASL technology for NASA's strategic exploration mission. The utility and fascination of foams are derived largely from the surprising fact that they have a solid-like elastic character in spite of being mostly gas with a few percent volume fraction of liquid, but can nevertheless flow under shear. The physical origin of such unusual rheology in terms of microscopic structure and dynamics is poorly understood and remains a subject of basic scientific interest to physicists, chemists, and chemical engineers. The proposed research promises important new insight into these issues, and could also have significant consequences for our understanding of flow in other dense randomly-packed systems such as emulsions, colloidal suspensions, slurries, bubbly liquids, and granular materials. Furthermore, all foam applications are empirically based and the proposed research may generate valuable fundamental guidance for the development of materials with more desirable rheology and better stability. The value of the FOAM project in terms of NASA’s exploration mission lies with Advanced Life Support. Water recovery and recycling are absolutely central to ASL, and require fine control of multiphase flows and separations. The very heart of the FOAM project is to bring new technology and understanding to these problems
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Foams are important for a clean environment. Firstly, their presence in natural waters signals the presence of surface-active impurities and a need for clean-up. Secondly, they are useful themselves in the cleanup process since many impurities can be removed by injecting gas and skimming off the resulting foam. Relatedly, study of foams brings specific insight into multiphase flow and separation phenomena of direct importance for water recovery and recycling operations.
Task Progress:	During this period my group included postdoc Chris Jones, Penn graduate student Adam Roth, UCLA graduate student Alex Gittings, and visiting summer intern Nicolas Fernandez. During the past year we have made significant progress on several different aspects of the overall FOAM project: First I helped finalize a new ESR (Experimental Requirements Document), where the focus shifted in the near term to coarsening. This is not a new direction, but rather represents a subset of the originally envisioned and funded ultimate microgravity experiments. The restricted experiment features optical probes, like DTS, DWS, and now SVS as crucial diagnostics of the structure and dynamics. In terms of communicating results, I gave invited presentations on foams at a summer school and at the Institute for Advanced Studies, plus my student, Adam Roth, gave two contributed talks and one poster presentation. Furthermore I wrote two peer-reviewed articles that have been accepted for publication. Research accomplishments fall into several categories. To begin, we made headway on a new rheology technique based on measuring the velocity profile in a channel caused by a known pressure drop. We were able to extract stress vs strain rate relations and fit to Herschel-Bulkley forms. The postdoc, Chris Jones, had to leave after writing up a preliminary draft. We succeeded in building a 2d sample cell in which the liquid fraction may be varied by controlling the filling depth of a surrounding trough of foaming solution. And we succeed in constructing software tools to extract bubble areas and side numbers and shapes, and to follow these quantities vs time. Furthermore we have constructed a model for the effects of inflation of the Plateau borders with liquid, and how this affects the coarsening rates for n-sided bubbles. This work will hopefully come to completion during the coming year. In addition we embarked upon several lower-dimensional coarsening experiments. In one, we introduce a wedge angle to the above 2d coarsening experiments. This breaks the symmetry, and introduces three-dimensional effects in a controlled way and hence offers important insights. In another we attempted to study coarsening between two hemispherical domes. Here the curvature of the embedding space alters the von Neumann law according to the area of a bubble -- so, for example, six sided bubbles may change area. We also are studying the final states for coarsening in confined geometries, where the system comes to complete equilibrium and ceases to evolve. This is being done in long cylindrical tubes, where "bamboo" structures are formed, as well as in hexagonal 2d cells, where hexagonal bubbles can tile the space. These experiences will be important as we look ahead to revisiting the analysis of the beautiful 3d tomography data we acquired, but were unable to fully analyze, during the prior period.
Bibliography Type:	Description: (Last Updated: 06/12/2025)

Abstracts for Journals and Proceedings	Roth AE, Durian DJ. "Coarsening in two-dimensional foams." March Meeting of the American Physical Society, Portland OR, March 2010. Bulletin of the American Physical Society 2010 Jan; 55:W12-9. , Jan-2010
Articles in Peer-reviewed Journals	Durian DJ, Raghavan SR. "Making a frothy shampoo or beer." Physics Today. 2010 May;63(5):62-3. http://dx.doi.org/10.1063/1.3431341 and see Full Text options. , May-2010
Books/Book Chapters	Dauchot O, Durian DJ, van Hecke M. "Dynamical heterogeneities in grains and foams." in "Dynamical heterogeneities in glasses, colloids, and granular materials." Ed. L. Cipelletti. to appear--expected publication December 2010., May-2010